Vidyalankar Institute of Technology

Semester V – INFT - Mid Semester Assessment – II

Date: 15/09/2025	Machine Learning	PC_PCC	20 Marks/ 45 Minutes

1 Solve any one (10 marks each)

A Apply Decision tree algorithm with Gini index using following dataset to decide the root node. — CO3

Loves Popcorn (X_1)	Loves Soda (X_2)	Age (X_3)	Loves Ice-cream (Y)
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

B Discuss Support machine vector algorithm in detail with its advantages and disadvantages. — CO3

2 Solve anyone (10 marks each)

A Explain the role of confusion matrix in evaluating classification models. — CO4

Problem statement: Suppose 10000 patients tested for flu; out of them 9000 are actually healthy and 1000 are actually sick. For the sick people, the test was positive for 620 and negative for 380. For the healthy people the same test was positive for 180 and negative for 8820.

Calculate Performance measures for the given problem statement for the following terms: 1. Accuracy, 2. Precision, 3. Recall, 4. F1 Score.

B What is Ensemble Learning? Explain it with Bagging, Boosting and Stacking Approaches. Explain the Random Forest algorithm with respect to its basic concept. — CO4

CO3	Apply appropriate machine learning algorithms to basic data-driven problems.
CO4	Analyze the performance and behavior of machine learning models using standard evaluation metrics.

Loves Popcorn	Loves Soda	Age	Loves Ice-cream
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
No	Yes	35	Yes
Yes	Yes	38	Yes
Yes	No	50	No
No	No	83	No

First column tells us whether or not someone loves popcorn, 2nd column whether or not they love soda, 3rd column their age, and whether or not they love Ice-cream.

We have to use this data to build this classification tree, which would predicts whether or not someone loves icecream.

How to build a tree start with just a data.

First thing we do is to decide the root node.

→ One of the popular method is Gini impurity, but there are also other methods like Gini impurity, i.e. Entropy Information Gain.

→ Numerically, the methods are quite similar, so we will focus on Gini impurity

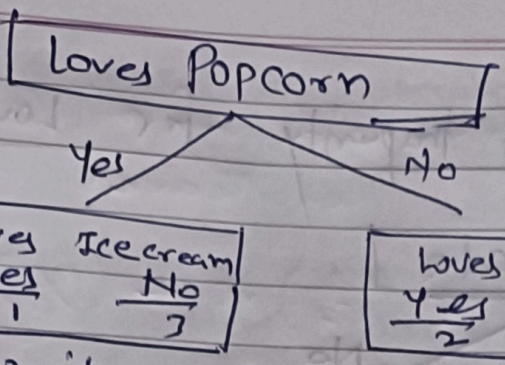
→ Let's start with Gini impurity with loves popcorn,
We start by calculating Gini impurity for individual leaves

Gini impurity for the leaf on left

$$\text{Gini Impurity} = 1 - (\text{the prob. of Yes})^2 - (\text{the prob of No})^2$$
$$= 1 - (1/4)^2 - (3/4)^2$$
$$= 1 - 0.0625 - 0.5625$$
$$= 0.375$$

Gini impurity for the leaf on right

Gini impurity = ~~$1 - 0.0625 - 0.5625$~~
$1 - (2/3)^2 - (1/3)^2$
= ~~0.375~~ $1 - (0.67)^2 - (0.33)^2$
$= 0.45$



Gini impurity = 0.375

Gini impurity = 0.444

Leaf on left have 4 people

leaf on right have only 3 people in it

The leaves do not represent same no. of people.

∴ Total Gini impurity = Highted avg. of Gini Impurities for the leaves.

Thus the total Gini Impurity is weighted avg. of the leaf impurities

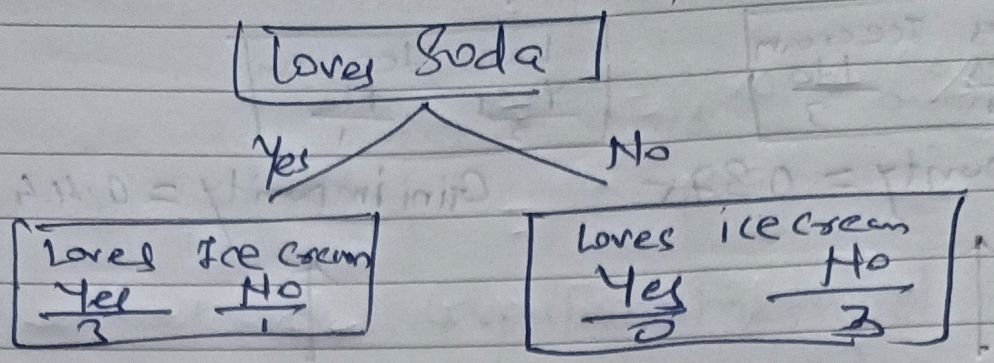
∴ Total Gini impurity = $\left(\frac{4}{4+3}\right) 0.375 + \left(\frac{3}{4+3}\right) 0.444$

$1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2$

$1 - 0.67 - 0.33 = 0.405$

∴ The Gini impurity for loves popcorn is 0.405

Like wise Gini impurity for Loves Soda



left leave

$$GI = 1 - (3/4)^2 - (1/4)^2$$

$$= 0.375$$

Right leaves

$$GI = 1 - (0/3)^2 - (3/3)^2$$

$$= 0$$

∴ ~~Avg.~~ Total Gini imp = wtd avg of Gini impurities of leaves

$$= \left(\frac{4}{7}\right) \times 0.375 + (3/7)\,0$$

$$= 0.214$$

Gini impurities for loves soda = 0.214

Now we need to calculate Gini impurity for age

Because Age contains numeric data, calculating Gini impurity involve more calculations.

→ The First thing we do is sort the rows by Age, from lowest value to highest val.

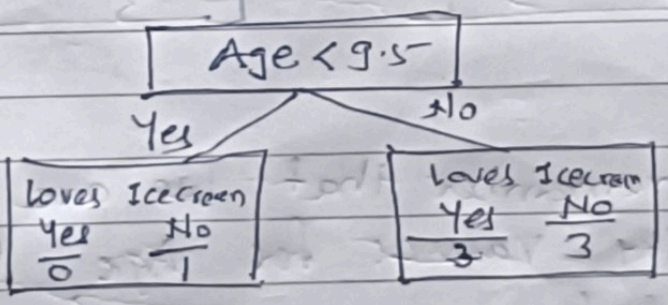
9.5 [7 GI = 0.429
12
15 [18 GI = 0.343
26.5 [35 GI = 0.476
36.5 [38 GI = 0.476
44 [50 GI = 0.343
66.5 [83 GI = 0.429

Calculate Avg Age of all adjacent people.

& then calculate Gini impurity values for each avg eg.

Ex ~~First avg~~. - to calculate Gini impurity

we put Age < 9.5 in the root



Gini impurity for leaf

On left	On Right
Impurity = $1 - \left(\frac{0}{0+1}\right)^2 - \left(\frac{1}{0+1}\right)^2$	Impurity = $1 - \left(\frac{3}{3+3}\right)^2 - \left(\frac{3}{3+3}\right)^2$
$= 1 - 1$	$= 1 - (0.5)^2 - (0.5)^2$
$= 0$	$= 0.5$

∴ Total Gini Impurity = Wtd avg of Gini impurity of leaves

$= (\frac{1}{7}) \times 0 + (\frac{6}{7}) \times 0.5$

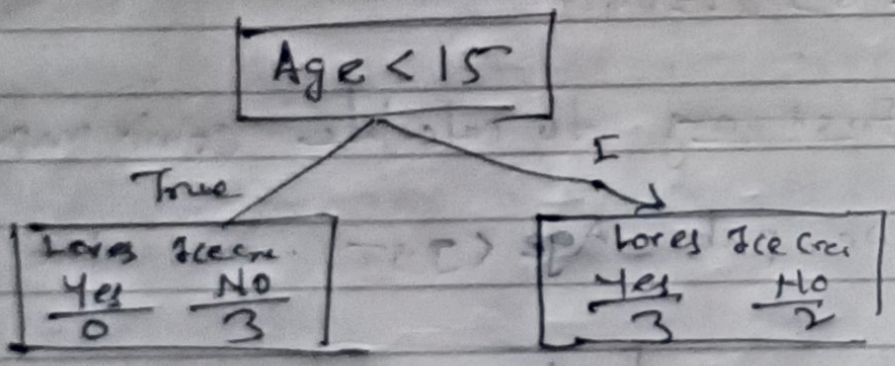
$= 0.429$.

So, we calculate Gini Impurity

9.5 → 0.429
15 → 0.343
26.5 → 0.476
36.5 → 0.476
54 → 0.343
66.5 → 0.429

15 & 54 are tied for the lowest impurity 0.343

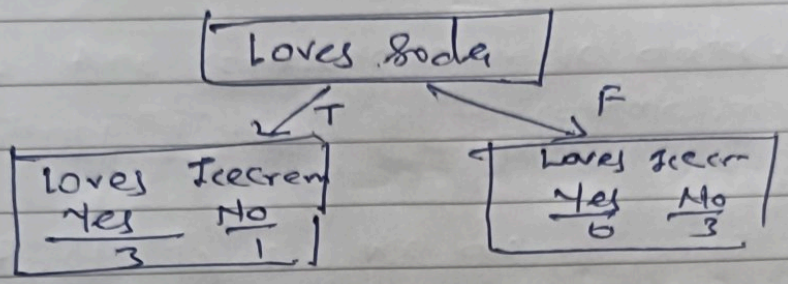
Either one.



However, remember that we are comparing Gini impurity values for Age, Loves popcorn & loves Soda.

Loves popcorn - 0.405
Loves Soda - 0.214
Age - 0.343.

which feature to be very top of the tree?
Loves Soda has the lowest Gini impurity overall.



Q.1B. Discuss support vector machine in detail with its advantages & disadvantages.

Support vector machine (SVM) is a supervised machine learning algorithm used for classification and regression tasks, though it is most commonly applied to classification problems.
The main idea behind SVM is to find the best decision boundary or hyperplane that separates datapoints of different classes with the maximum margin.

Working of SVM

1. Linear Separation.
   - For linearly separable data, SVM finds a straight line (2D), plane (3D), or hyperplane (in higher dimensions) that maximizes the distance between the classes. (margin)
   - The datapoints closest to hyperplane or str. line are called as Support vectors.
2. Maximizing the margin:
   The objective is to maximize the dist betn the hyperplane & support vectors. A larger margin reduces classification error.
3. When data is not linearly separable, SVM uses kernel functions, e.g. polynomial, RBF or Sigmoid to map the data into a higher dimensional space where a linear separation is possible. This is called the kernel ~~trick~~ tricks.

Advantages of SVM:

- Effective in high dimensions.
  Work even when no. of samples is less than number of features (attributes).
  Ex. text classification (spam detn & Sentiment Analysis)
- Robust to overfitting. (maximizing 'margin')
- Can handle linear & non-linear classification problems by choosing appropriate kernel functions.
- Clear margin of separation.
- Works well with small to medium datasets.

Disadvantages of SVM

- Computationally expensive Training complexity $O(n^2)$ or worseer especially with large datasets.
- Incorrect kernel selection can lead to poor performance.
- Not suitable for very large datasets.
- Poor performance on noisy data.

Applications of SVM

- Text & hypertext classification (spam filter, sentiment Analysis)
- Handwritten digit recognition
- Bioinformatics (protein classification, cancer detection)
- Image classification & facial expression recognition
- Fraud detection & anomaly detection

Q2A. Explain the role of confusion matrix in evaluating classification models.

- A confusion matrix is a tabular representation ~~of~~ used to evaluate the performance of classification model.
- It ~~pr~~ compares predicted class labels with actual (true) class labels, providing a complete picture of how well the model is performing.

For the binary classification problems the structure of confusion matrix is

	Predicted +ve	Predicted -ve
Actual +ve	TP	FN
Actual -ve	FP	TN

Role of confusion matrix in Evaluation

- Accuracy: Proportion of correctly classified instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

- Precision: How many predicted +ves are actually correct

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Recall: How many +ves were correctly identified

$$\text{Recall} = \frac{TP}{TP + FN}$$

- F1 score — Harmonic mean of precision & recall.

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

-> Let's have a confusion matrix

	Predicted Positive	Predicted -ve
Actual sicks	620	380
Actual Healthy	180	8820

For sick people (Actual Positive = 1000)

+ve test -> TP = 620

-ve test -> FN = 380

For Healty people (Actual -ve = 9000)

+ve test = 180 -> FP = 180

-ve tels = 8820 -> TN = 8820

DATE / /

(1) $\therefore$ Accuracy $= \frac{TP + TN}{TP + FP + FN + TN}$

$= \frac{620 + 8820}{620 + 380 + 180 + 8820} = 0.944$

(2) Precision $= \frac{TP}{TP + FP}$ — Predictions

$= 0.775$

(3) Recall $= \frac{TP}{TP + FN} = \frac{620}{380 + 620} = 0.62$

Truth

(4) F Score $= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

$= 2 \times \frac{0.775 \times 0.62}{0.775 + 0.62}$

$= 0.689$

$= 68.9\%$

Accuracy = 94.4%
Precision = 77.5%
Recall = 62%
F1 Score = 68.9%

Q2B. What is ensemble learning?

Ensemble learning is a machine learning technique where multiple models are combined to form a strong learner. Instead of relying on a single model ensemble learning aggregates predictions from multiple models to improve accuracy, robustness & generalization.

Following are the ensemble learning Approaches:

1) Bagging
2) Boosting
3) Stacking.

1) Bagging (Bootstrap Aggregating)

It reduces variance by training models in parallel on different random subset of data.

The process involves the following points/steps:

1. generate multiple bootstrap samples.
2. Train a base model, often Decision Trees on each sample.
3. Aggregate predictions, i.e. majority votes for classification & avg. for regression.

Examples for Bagging - i) Random Forest.

Advantages: Reduces variance & prevent overfitting.

Disadvantage: May not reduce Bias significantly.

2) Boosting

It reduces bias by training models sequentially where each model focuses on correcting errors made by the previous ones.

The process involves the following steps

- Train a weak learner on data
- Identify misclassified samples & give them higher weights.
- Train the next learner to focus on these difficult cases
- Combine learners with weighted voting.

Examples are AdaBoost, Gradient Boosting, XGBoost.
Advantage - very powerful, reduces both bias & variance.
Disadvantages: Sensitive to noise, risk of overfitting if not tunned.

3) Stacking:
It combines multiple different models & use a meta learner to make the final prediction.
The process involves following steps.
1. Train different base learners, i.e. decision tree, logistic regression, SVM
2. Collect predictions of base learner as features.
3. Train a meta-model often logistic regression on these features for classification problems.

Example - kaggle competition winners often use stacking

Advantage: Can capture complex relationship by leveraging strengths of different models.
Disadvantage:
More complex, requires careful validation to avoid overfitting.

Random Forest Algorithm

Random Forest is an ensemble algorithm based on bagging with additional randomness. It builds forest of decision trees, each trained on random samples of data & random subset of features.

It involves following steps,

1. Create multiple bootstrap samples from the dataset.
2. For each tree
   - Randomly select subset of features at each split.
   - Grow a decision tree.
3. Aggregate predictions,
   - Classification - majority votes
   - Regression avg. outputs.

Let us consider the following dataset with 4 samples to build & to create a Random Forest

Chest Pain	Good Blood Circulatn	Blocked Arteries	Weight of Patient	Heart Disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	Yes	167	Yes

Step 1.

Let us create a first Bootstrap sample to build a decision tree

Chest Pain	Good Blood Circulation	Blocked Arteries	Wt of Patient	Heart Disease
Yes	Yes	Yes	180	Yes
No	No	No	125	No
Yes	No	Yes	167	Yes
Yes	No	Yes	167	Yes

One We ignores one sample from the original dataset & replace with duplicate sample (here we ignore third sample & taken 4th sample twice.

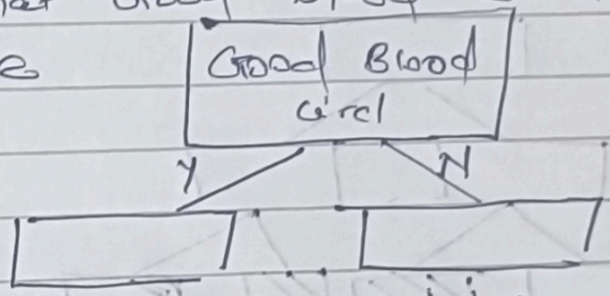
Step 2

Create decision tree using this bootstrapped dataset, but by using random subset of features, i.e. columns

In this example, we will consider 2 variables at each step to build a decision tree.

We will select Good Blood Circulation & Blocked Arteries as a candidate for root node.

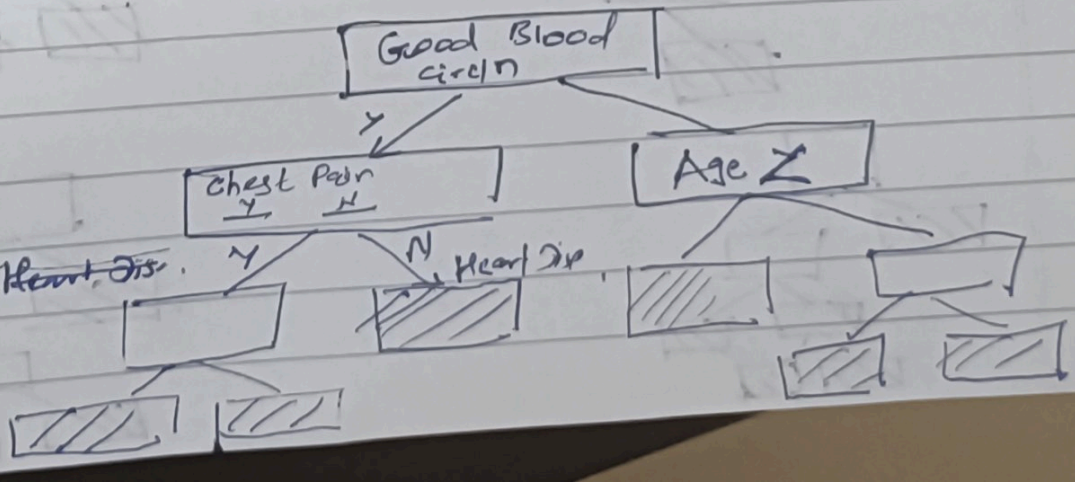
Let us assume that Good Blood circulation as the root node



Now, we will focus on remaining 3 i.e, chest pain, Blocked Arteries & wt of patient.

- From Blocked Arteries & Chest pain we select- Chest pain,
- Then Blocked Arteries & Age we select Age.

And finaly Blocked Arteries we get the following Decision Tree.



So we have built a decision tree using bootstrapped dataset, by considering only a random subset of variables.

Now, we go back to step 1, we will make a new bootstrap dataset & build a tree considering subset of variables at each step. and repeated this 5 times, 10 time's 100 times & so on depending on the size of dataset.

Let us So finally we have created a random forest of ~~fotto~~ of our original dataset with following 6 decision trees.

